

TOPIC 4 week 6 test 2

Given a sample of 8 student's score in test 1 as follows: **ungrouped data - sample**

78 80 80 70 45 50 98 23

Find the

(a) mean (b) mode (c) median (d) range (e) variance (f) standard deviation

(a) $\bar{x} = \frac{\sum x}{n}$ (b) 80 (c) 23, 45, 50, 70, 78, 80, 98 position = $\frac{8+1}{2} = 4.5^{th}$

$\bar{x} = 524/8 = 65.5$

= 74

(d) = 98 - 23 = 75

(e) $s^2 = \frac{\sum x^2 - \frac{(\sum x)^2}{n}}{n-1}$ (f) 24.26

$$= \frac{38442 - \frac{(524)^2}{8}}{7} = \frac{4120}{7} = 588.57$$

The following table shows the number of letters per word for 30 selected words taken from a local novel.

grouped data - sample

Number of Letters	Frequency	midpoint, m	mf	m ² f
10 - 12	2	11	22	242
12 - 14	3	13	39	4563
14 - 16	4	15	60	900
16 - 18	5	17	85	1445
18 - 20	7	19	133	2527
20 - 22	4	21	84	1764
22 - 24	3	23	69	1587
24 - 26	2	25	50	1250

542 10 222

(a) mean (b) mode (c) median (d) variance (e) standard deviation

midpoint = $\frac{L.B + U.B}{2}$ / $\frac{L.L + U.L}{2}$

(a) mean, $\bar{x} = \frac{\sum mf}{\sum f} = \frac{542}{30} = 18.07$

(b) mode = step 1: mode class

step 2: mode = $L + \left[\frac{f_m - f_B}{(f_m - f_A) + (f_m - f_B)} \right] C$

mode class = 18-20 [class boundary]

mode = 18 + $\left[\frac{7 - 5}{(7 - 4) + (7 - 5)} \right] [20 - 18]$

mode = 18 + $\left[\left(\frac{2}{2} \right) (2) \right] = 18.8$

(c) median

step 1 : position = $\Sigma f / 2$ position = 15th

step 2 : median class

step 3 : median = $L + \left(\frac{\frac{\Sigma f}{2} - F_L}{f_m} \right) C$

median class

= 18-20 (class boundary)

$$18 + \left(\frac{\frac{30}{2} - 14}{7} \right) 2$$

$$= 18.29$$

median ← f_m ← lower/before

$$(d) \text{ variance, } s^2 = \frac{\Sigma m^2 f - \frac{(\Sigma m f)^2}{\Sigma f}}{\Sigma f - 1} = \frac{10222 - \frac{(542)^2}{30}}{29} = 14.82$$

(e) std. dev = 3.85

The following table shows the petrol consumption (RM per week) of 30 families.

class boundary	Class	Frequency	midpoint	$m f$	$m^2 f$
0.5 - 49.5	0 - 49	1	24.5	24.5	600.25
49.5 - 99.5	50 - 99	5	74.5	372.5	27751.3
99.5 - 149.5	100 - 149	6	124.5	747	93001.5
149.5 - 199.5	150 - 199	11	174.5	1919.5	
199.5 - 249.5	200 - 249	5	224.5	1122.5	
249.5 - 299.5	250 - 299	2	274.5	549	

30

4735

659008

$$(a) \text{ mean } \bar{x} = 4735/30$$

$$= 157.83$$

(b) mode class = 150-199

$$L = \left[\frac{f_m - f_b}{(f_m - f_b) + (f_m - f_a)} \right] C$$

$$= 149.5 + \left[\frac{11 - 6}{(11 - 5) + (11 - 6)} \right] 50$$

$$= 170.23$$

(d) median position

$$30/2 = 15th$$

median class = 150-199

$$149.5 + \left(\frac{30/2 - 12}{11} \right) 50$$

$$= 163.14$$

$$(e) \text{ variance: } s^2 = \frac{\Sigma m^2 f - \frac{(\Sigma m f)^2}{\Sigma f}}{\Sigma f - 1}$$

$$= \frac{659008 - \frac{(4735)^2}{30}}{29}$$

$$= 3850.60$$

$$(f) \text{ std. dev } = 62.05$$